

Scalability & Future Plan

Date	18 JULY 2026
Team ID	SWTID-2026-4144
Project Name	RISING WATERS
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited to single database server	Can slow down with more users/data	High
2	Manual data entry for water levels	Increases human error&delay updates	Medium
3	Basic security , no role-based access	Risk of unauthorized access to data	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports 50-100 users	Use cloud hosting +load balancer to support 10k+users
2	Data Storage	Local MYSQL database	Migrate to scalable cloud DB like firebase/AWS RDS
3	Performance	Single-threaded processing	Implement caching with Redis+optimize queries
4	Security	Basic login system	Add JWT authentication,HTTPS,and role-based access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	IOT sensor integration for real time water level monitoring	2-3 months	Removes manual entry,gives live accurate data
Phase 3	Mobile app with push notifications & alerts	4-6 months	Users get instant alerts for low/high water levels, better accessibility
Phase 4	AI/ML for water usage prediction + Analytics dashboard	8-12 months	Helps in forecasting demand , reduces wastage , supports decision making